SHOP PERFORMANCE ANALYSIS

PRESENTED BY
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1.INTRODUCTION OVERVIEW

The objective of this project is to analyse the online shop's performance from January 2024 to June 2026 by examining revenue, orders, products, customers, payments and discounts. The analysis aims to identify key trends, business performance and opportunities for improvement.

TOOLS USED:

Python | Pandas | Databricks | Looker Studio

OVERALL BUSINESS PERFORMANCE:

- The shop generated total actual revenue of 2,990,114.95 from 42,721 successful orders, with an average order value of 69.99.

EXECUTIVE SUMMARY:

- Electronics generated the highest revenue, while Accessories recorded the highest number of units sold.
- Revenue declined during the first half of 2026, highlighting the need to investigate sales performance and improve business strategies.

2.DATA CLEANING & PREPARATION

The analysis used four datasets:

- Customers – customer age, city, signup date and customer segment.
- Orders – order date, products, quantity, discounts, payment methods and order status.
- Products – product names, categories and unit prices.
- Payments – payment dates and payment status.

DATA CLEANING

- Checked all datasets for missing values and duplicates.
- Removed exact duplicate order records.
- Removed records with missing order dates and quantities.
- Removed invalid quantities of zero or below.
- Filled missing discounts with 0.
- Replaced missing payment methods with "Unknown".
- Standardised inconsistent city names such as "tehran" to "Tehran" and "Mashad" to "Mashhad".
- Filled missing customer ages using the median.
- Checked broken links between customers, orders, products and payments.

DATA INTEGRATION

Orders were joined with Products, Customers and Payments to create one final dataset for analysis.

The final dataset contained 49,860 rows.

REVENUE RULE

Revenue = Quantity × Unit Price × (1 − Discount)

Only orders that were both Completed and Paid were counted as Actual Revenue. Cancelled, Returned, Failed and Refunded transactions were excluded from Actual Revenue.

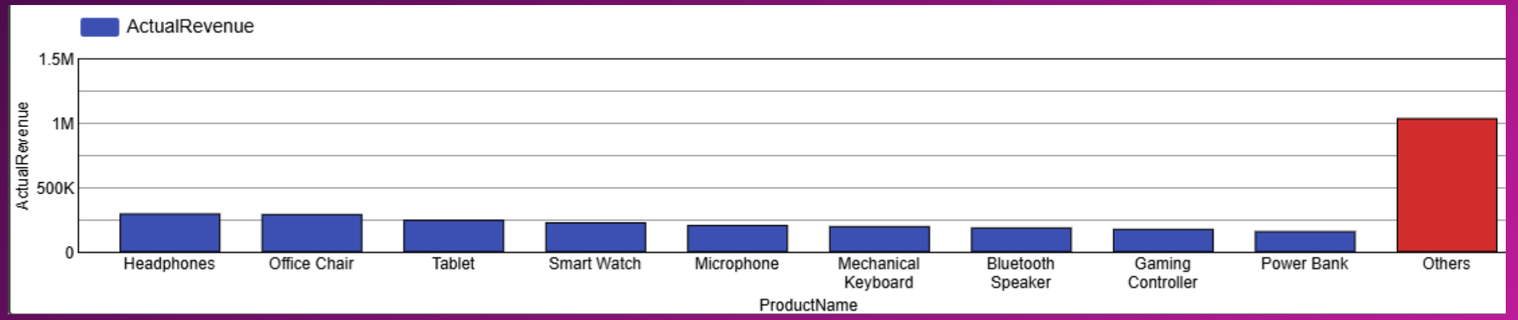
3.PRODUCT PERFORMANCE

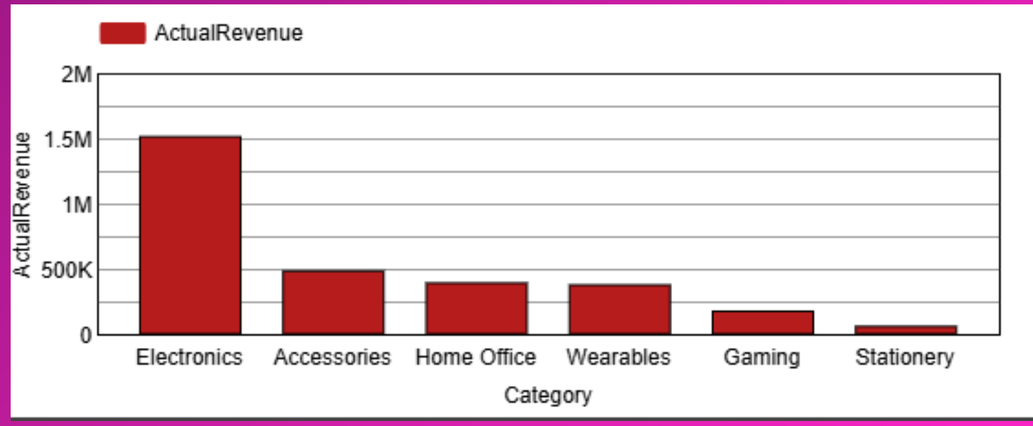
PRODUCT PERFORMANCE-KEY INSIGHTS

- Headphones generated the highest revenue of 292,207.50.
- Electronics was the highest-revenue category, generating 1,511,908.80.
- Accessories recorded the highest sales volume, with 30,013 units sold.

BUSINESS INSIGHTS:

High revenue does not always mean high sales volume. The business should focus on profitable product categories while maintaining stock of popular items.

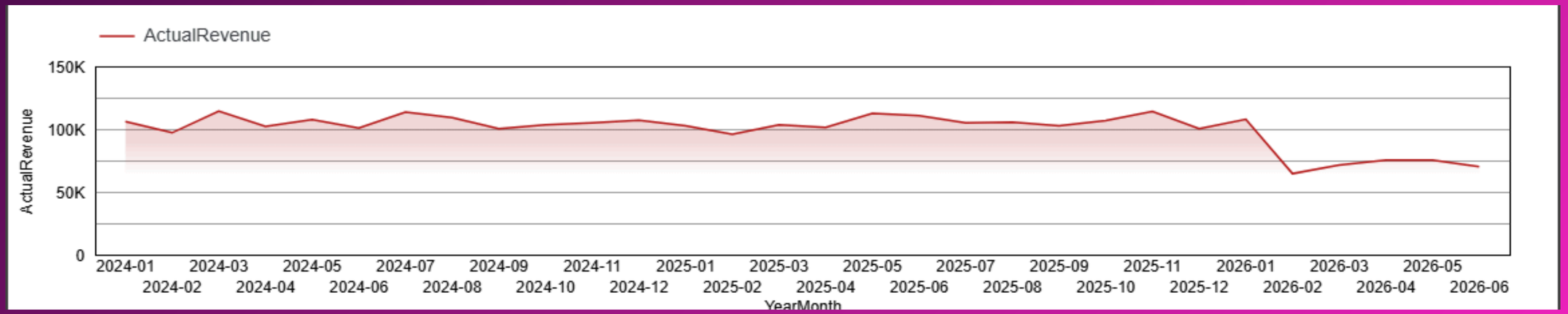




4.MONTHLY REVENUE TREND

- Revenue remained relatively stable throughout 2024 and 2025.
- March 2024 recorded the highest monthly actual revenue at 114,176.80.
- February 2026 recorded the lowest monthly actual revenue at 64,512.90.
- Revenue declined noticeably during the first half of 2026.

KEY INSIGHT:
The business maintained relatively stable revenue during 2024 and 2025, but the decline in early 2026 should be investigated to understand what caused the change.

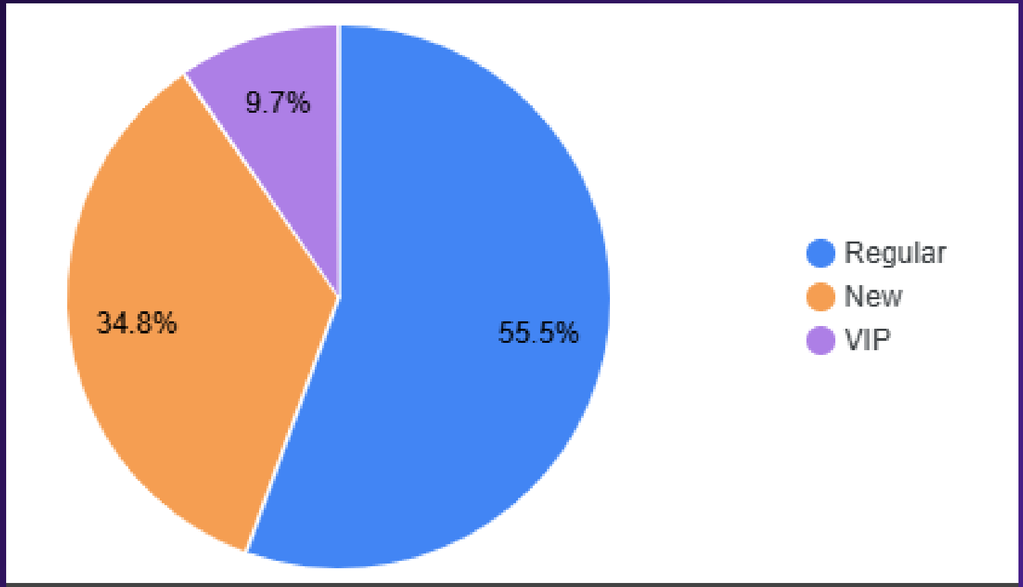


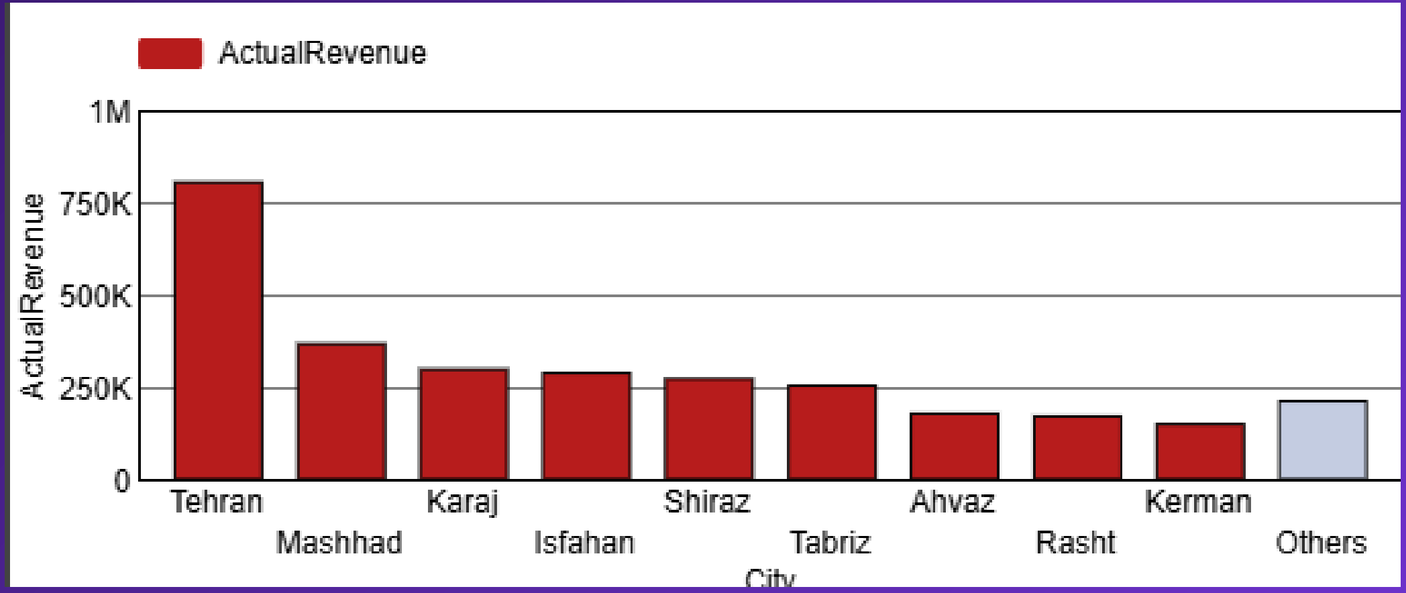
5.CUSTOMER PERFORMANCE

- Tehran generated the highest actual revenue at 822,877.65.
- Mashhad was the second-highest performing city with 376,771.00.
- Regular customers were the highest-value customer segment, generating 1,659,781.55.
- New customers generated 1,040,078.80, while VIP customers generated 288,519.45.

KEY INSIGHT:

Tehran is the strongest revenue-generating city, while Regular customers contribute the largest share of customer revenue.





6.OPERATIONAL PERFORMANCE

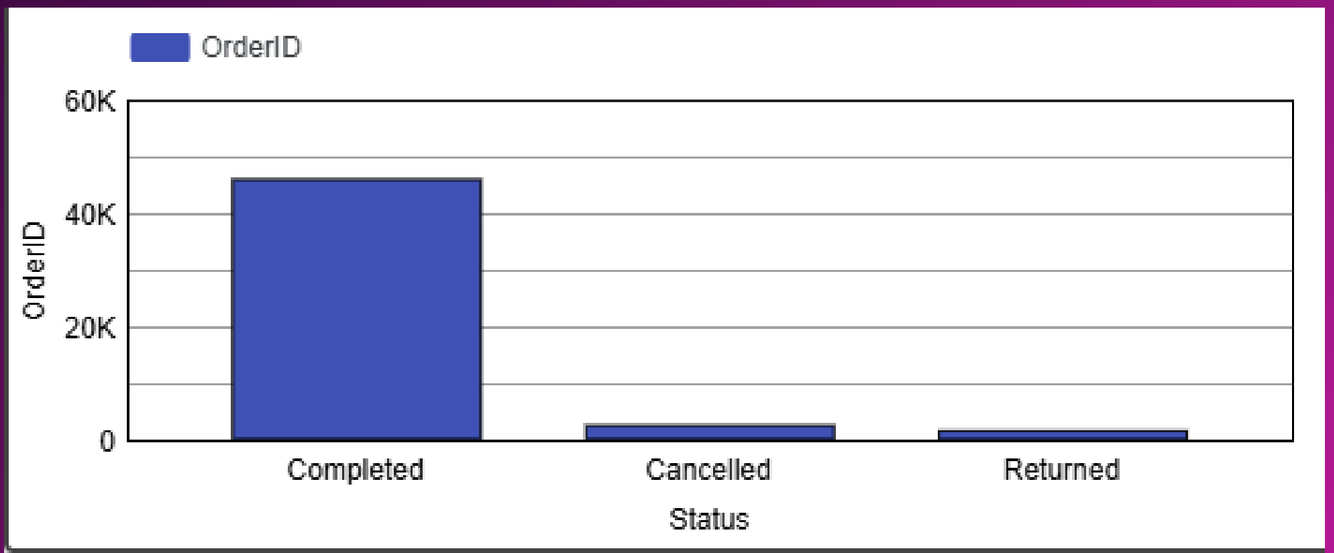
Objective: To evaluate order completion, cancellations, returns, and payment failures.

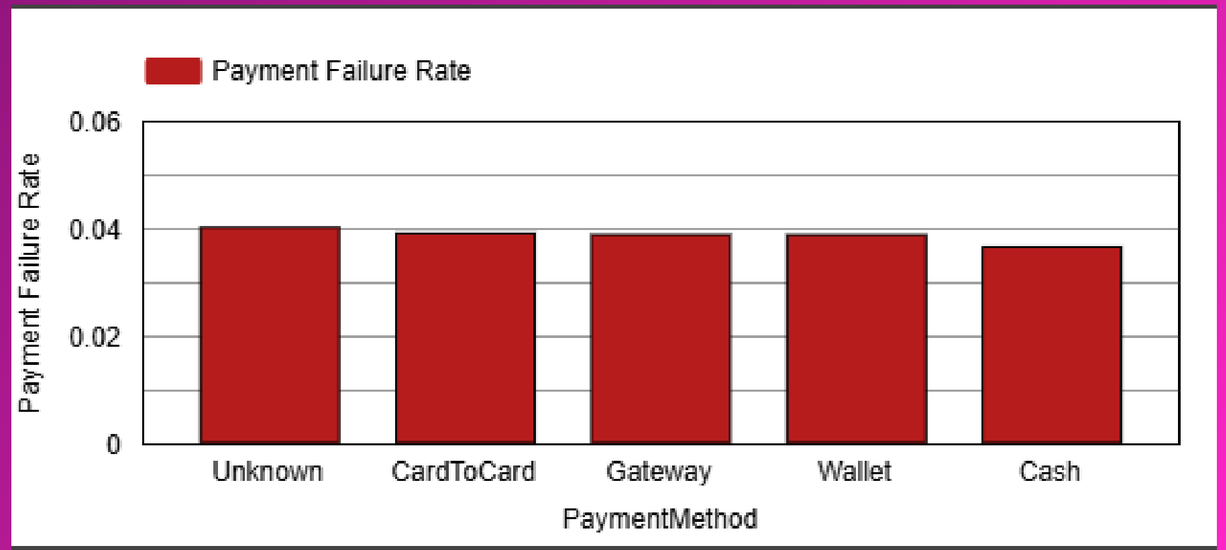
KEY INSIGHTS:

- The shop recorded 45,867 completed orders, 2,463 cancelled orders, and 1,530 returned orders.
- Cancelled and returned orders accounted for 8.01% of total orders.
- A total of 42,721 orders were successfully completed and paid.
- The overall payment failure rate was 3.85%.
- Payment failure rates varied slightly across payment methods, highlighting opportunities to improve payment processing.

BUSINESS INSIGHTS:

The majority of orders were completed successfully. However, cancellations, returns, and failed payments remain areas where the business can improve operational efficiency.





7.DISCOUNT ANALYSIS

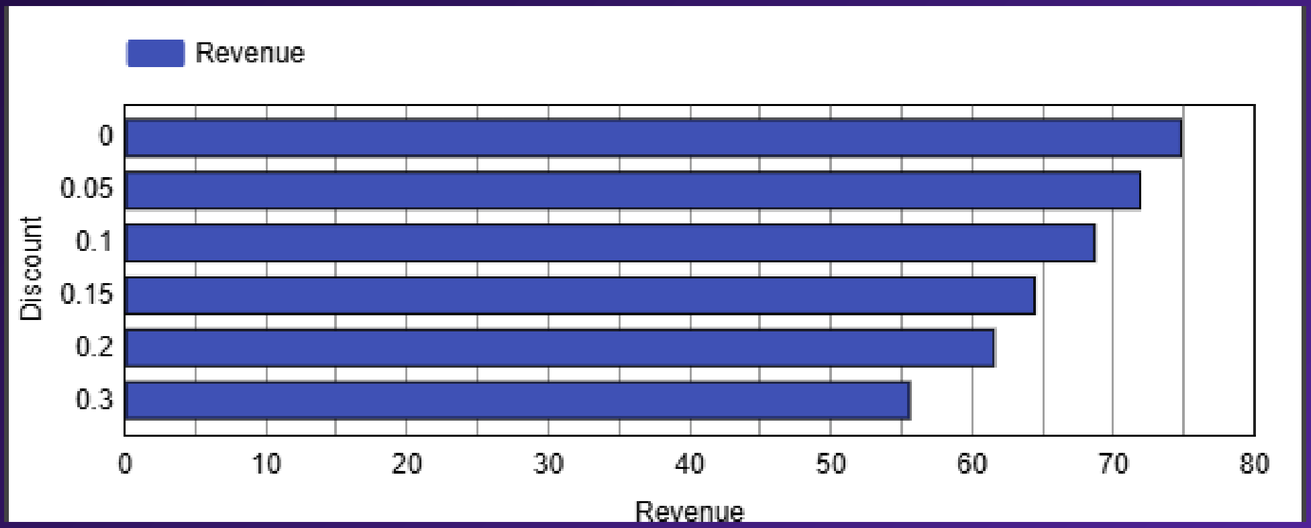
understanding how discounts affect customer purchases and revenue.

KEY INSIGHTS:

- The shop offered discounts ranging from 0% to 30%.
- Higher discounts did not significantly increase the number of units purchased.
- Average revenue decreased as discount levels increased.
- Larger discounts did not necessarily result in better sales performance.

BUSINESS INSIGHTS:

The business should review its discount strategy and focus on targeted promotions rather than offering large discounts that may reduce revenue.



8.BUSINESS RECOMMENDATIONS

Based on the Shop Performance Analysis, the following recommendations are proposed:

1. Prioritise High-Performing Products

- Focus on high-revenue products such as Headphones, Office Chairs, and Tablets.
- Promote Electronics and use popular Accessories for cross-selling opportunities.

2. Improve the Discount Strategy

- Avoid offering large discounts that reduce average revenue.
- Introduce targeted promotions to encourage purchases while protecting revenue.

3. Investigate the Revenue Decline

- Investigate the decline in revenue during the first half of 2026.
- Improve order processing and payment systems to reduce cancellations, returns, and failed payments.

DATA QUALITY NOTES:

The analysis was based on cleaned data from four datasets. Missing values, duplicates, invalid quantities, and inconsistent records were addressed before analysis to improve data reliability.